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EXPERIMENT - IIA

ROUTING AT NETWORK LAYER

AIM:

To simulate & configure static routing between multiple networks using cisco packets tracers, configure primary & backup static routes using administrative distance.

REQUIREMENTS:

* Cisco Packet Tracer
* Three router - router 0, 1, 2
* PCs/end devices
* properly assigned IP address.

OBSERVATION:

→ A network topology consisting of 3 routers, Routers 1, 2 was created in cisco packet tracer, different networks were connected & were assigned suitable IP address.

Router 1:

→ was configured with static routes to the 10.0.0.0/8 to 40.0.0/8 networks. The router through Router 0 was assigned a lower administrative distance primary route.

Router 2:

→ Router 2 was configured with static router to the 10.0.0/8 & 30.0.0.0/8 networks. The routing table & verified using show ip route static.

Backup Route Verification:

→ The main link b/w Router 0 & Router 1,2 was disconnected to simulate a link failure connectivity was verified using ping & tracer: confirming that the backup route was working successfully.

Verification:

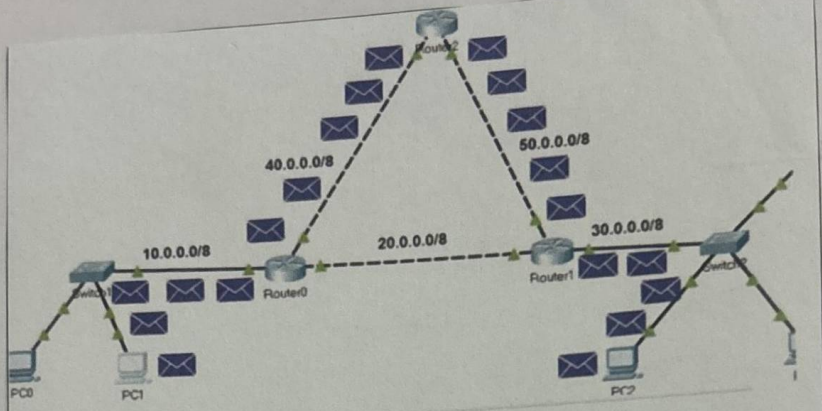
→ Show ip static route. The routing table showed the configured static routes & their Administrative using the ping command successfully ping between source & destination networks.

Deleting a static route:

→ Show ip static route. After deleting the main route, the backup route becomes the active if the alternate path is available.

RESULT:

Thus, static routing was successfully configured and simulated using cisco packet tracer. The primary & backup routes were configured using different ways.



Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>enable
Router#show ip route static
     30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
S       30.0.0.0/8 [10/0] via 20.0.0.2
S       30.0.0.100/32 [10/0] via 40.0.0.2
S    50.0.0.0/8 [10/0] via 40.0.0.2

Router#
```

PC1

Physical Config Desktop Programming Attributes

```
Packet Tracer PC Command Line 1.0
C:\>tracert 30.0.0.100

Tracing route to 30.0.0.100 over a maximum of 30
hops:

  1    1 ms       0 ms       0 ms       10.0.0.1
  2    *          0 ms       0 ms       40.0.0.2
  3    *          *          0 ms       20.0.0.2
  4    *          0 ms       0 ms       30.0.0.100

Trace complete.

C:\>
```

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EXPERIMENT - 11B

SIMULATE RIP USING CISCO PACKET TRACES

AIM:

To simulate RIP (exciting information protocol) using cisco Packet Tracer & verify dynamic routing between routers.

PROCEDURE:

→ Create the network topology with 3 routers & 2 PCs. in Cisco packet tracer.
→ Assign IP addresses to the PCs to router interfaces.
→ Enable the router in the interface.
→ configure clock rate & based on the DHCP side of serial link.
→ Enable RIP using the router rip command.
→ Advertise the directly connected network using network command.
→ Use trace to verify the router selected by RIP.
→ Disconnect the Primary Table.

PROGRAM:

Router 0:

```
enable
configure terminal
interface fastethernet 0
ip address 192.168.1.249.255.235.235
clock rate 64000
bandwidth 64
shutdown
exit
```

interface serial 0/01
ip address 1912.168.1.254.255.255.28
clock rate 64000
bandwidth 64
no shutdown
exit

router trip
network 100.0.0
network 192.168.1.248
network 192.168.1.233
exit

Router 1:

enable
config terminal
interface serial 0/0/0
ip address 192.161.1.250.255.255.236
no shutdown
exit
interface serial 0/0/1
ip address 192.168.1.248.256.255.255
clock rate 64000
no shutdown
exit

route trip
network 192.168.1.248
network 192.168.1.244
exit

Router 2:

enable
configure terminal
interface fastethernet 0/1
ip address 20.0.0.1.255.0.0.0
no shutdown
exit

interface serial 1/0/1
ip address: 192.168.1.245.255.255.255
no shutdown
exit

interface serial 0/0/1
ip address 192.168.1.256.255.255.255
no shutdown
exit

PC0:

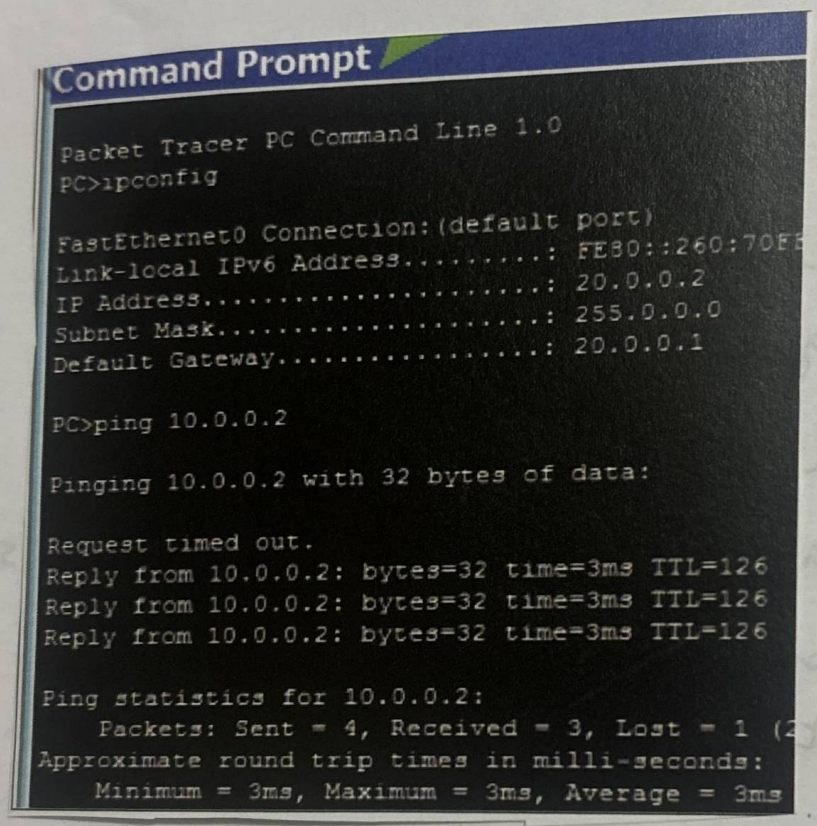
IP address: 10.0.0.2
subnet mask: 255.0.0.0
default gateway: 10.0.0.1

PC1:

IP address: 210.0.0.2
subnet mask: 255.0.0.0
default gateway: 20.0.0.1

Testing:

Ping: 20.0.0.1
Tracert: 20.0.0.2


Command Prompt
Packet Tracer PC Command Line 1.0
PC>ipconfig
FastEthernet0 Connection:(default port)
Link-local IPv6 Address.....: FE80::260:70FE
IP Address.....: 20.0.0.2
Subnet Mask.....: 255.0.0.0
Default Gateway.....: 20.0.0.1
PC>ping 10.0.0.2
Pinging 10.0.0.2 with 32 bytes of data:
Request timed out.
Reply from 10.0.0.2: bytes=32 time=3ms TTL=126
Reply from 10.0.0.2: bytes=32 time=3ms TTL=126
Reply from 10.0.0.2: bytes=32 time=3ms TTL=126
Ping statistics for 10.0.0.2:
Packets: Sent = 4, Received = 3, Lost = 1 (
Approximate round trip times in milli-seconds:
Minimum = 3ms, Maximum = 3ms, Average = 3ms

RESULT:

Thus, the RIP was successfully configured & simulated using Cisco Packet tracer connectivity between PC0 & PC1 was written & verified & RIP automatically selected & altered, when the primary router failed.